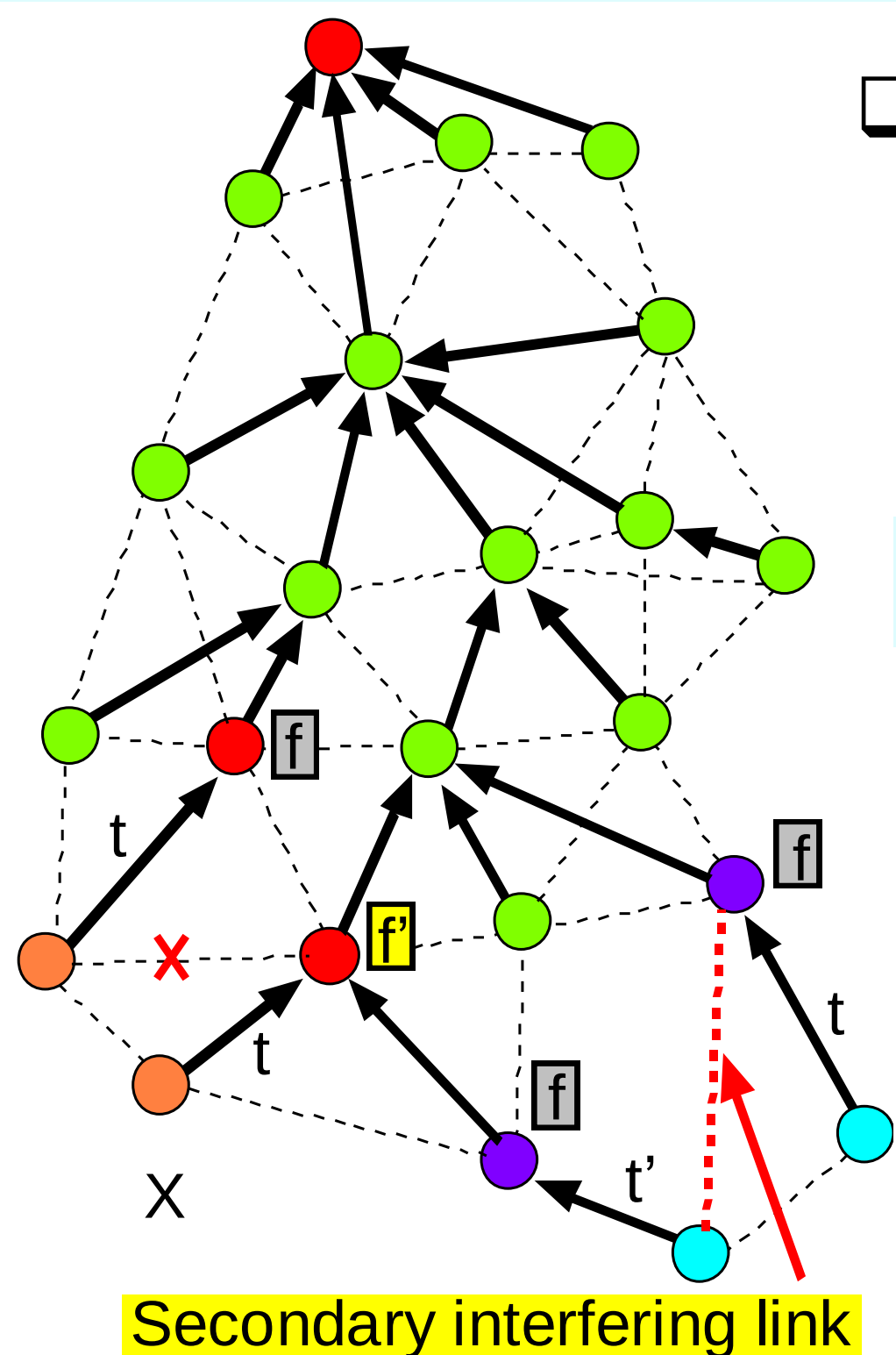


Bounded-Degree Minimum-Radius Spanning Trees for Fast Data Collection in Sensor Networks

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Motivation

Maximize Aggregated Sink Throughput + Minimize Delay



Applications

- High data-rate mission critical (e.g., SHM)
- Real-time continuous monitoring (e.g., wildfire, security-surveillance, environmental)

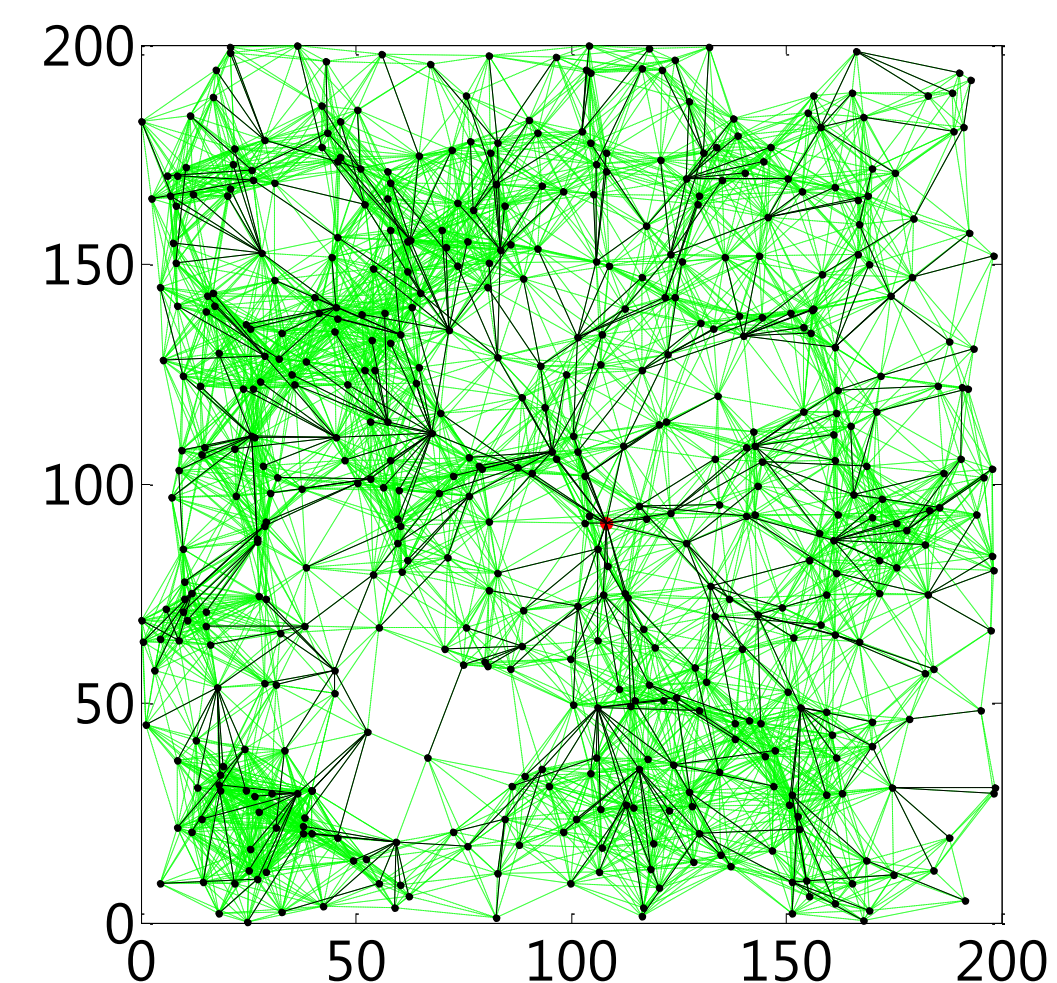
Challenges: Scheduling and Routing

TDMA Scheduling

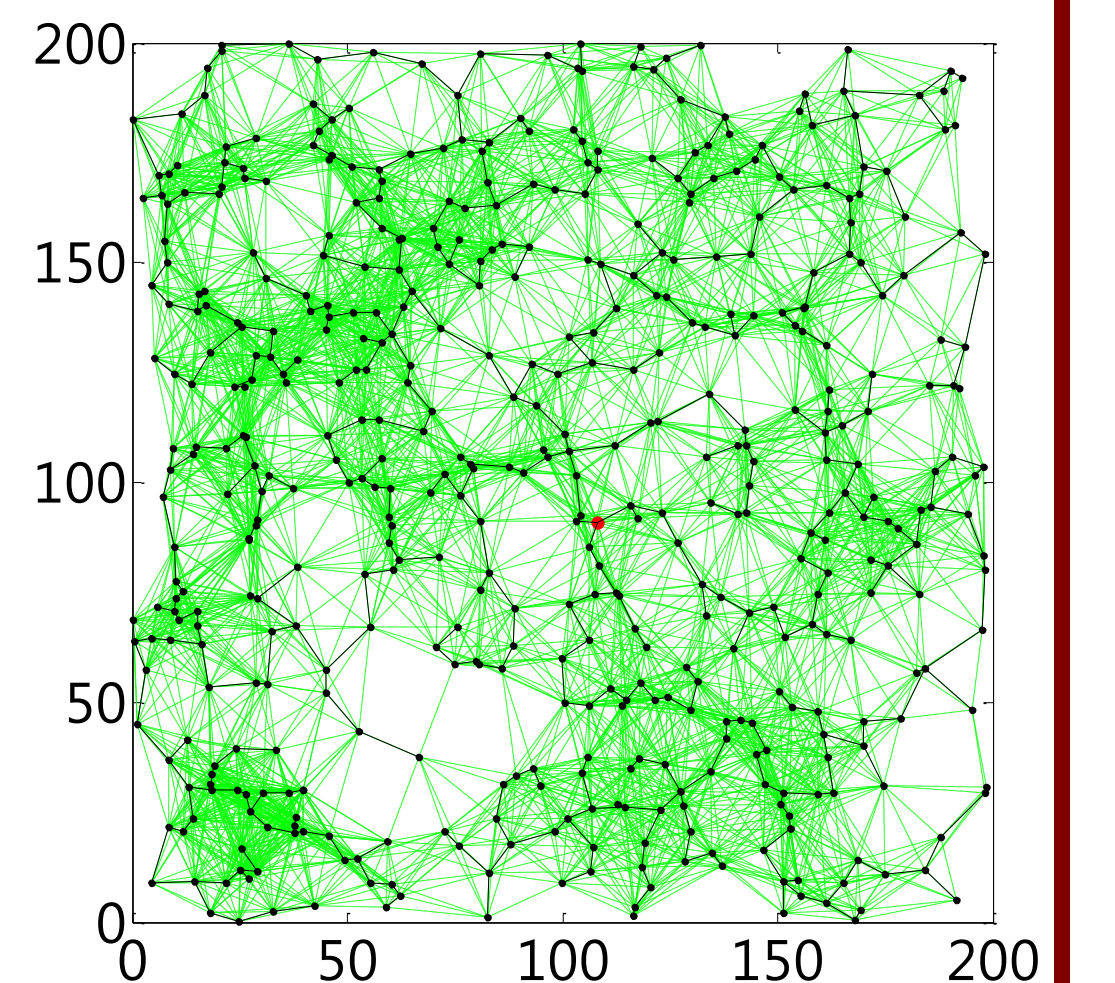
- Half-duplex transceivers (primary conflict)
- 2-hop interfering links (secondary conflict)

Routing Topology

- High node degree + small radius (SPT) => fewer concurrent transmissions but low delay
- Low node degree + large radius (MIT) => more concurrent transmissions but high delay



Shortest Path Tree (SPT):
high node degree, small radius



Minimum Interference Tree (MIT): low node degree, large radius

Problem Formulation and Solution

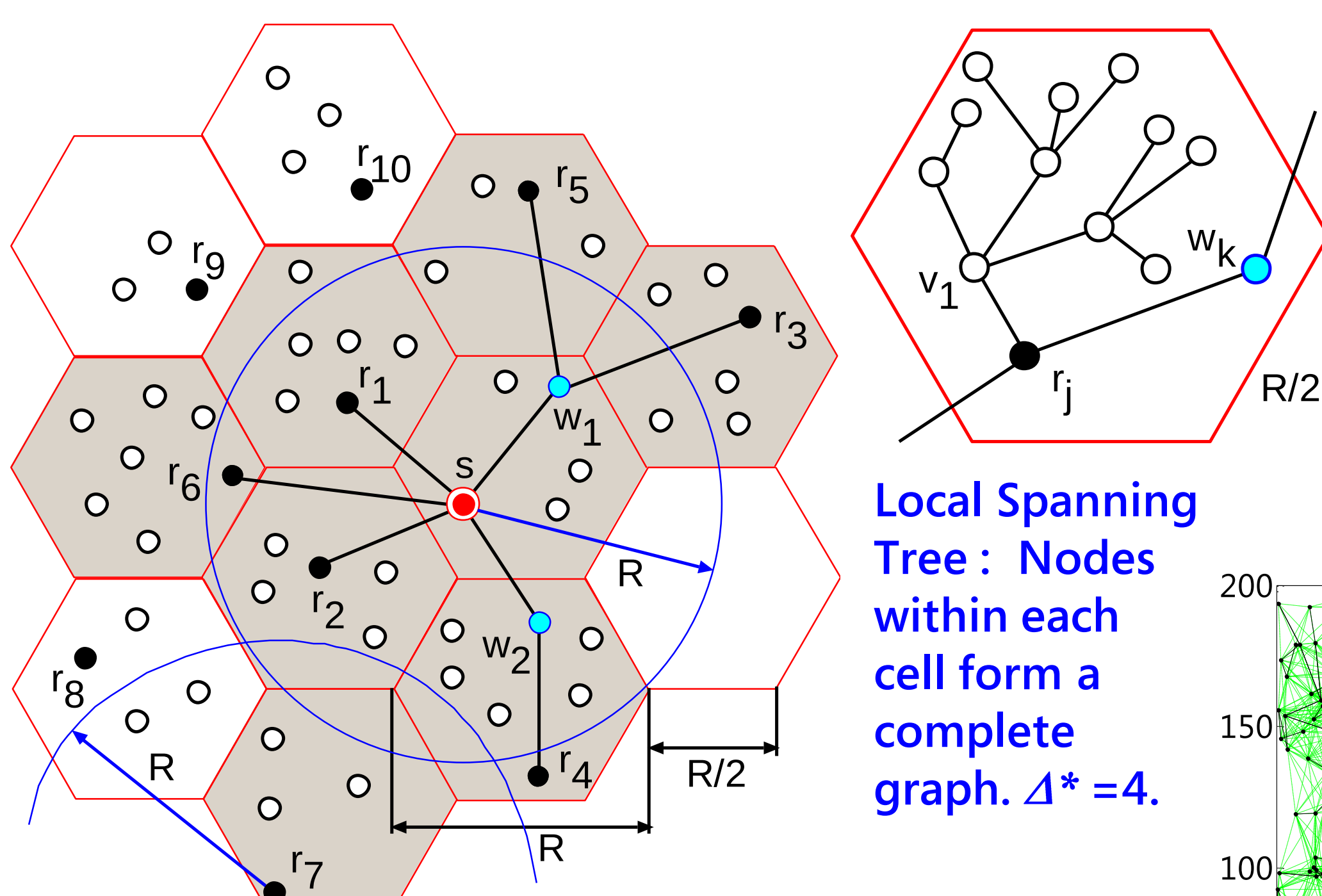
Bi-Criteria Optimization Formulation

Bounded-Degree Minimum-Radius Spanning Tree (NP-Hard)

- Given a graph $G = (V, E)$, communication range R , and a constant $\Delta^* \geq 2$, construct a spanning tree T of G rooted at sink $s \in V$, such that:
 - Radius (max hop count to sink) of T is minimized
 - Degree of any node in T is at most Δ^*

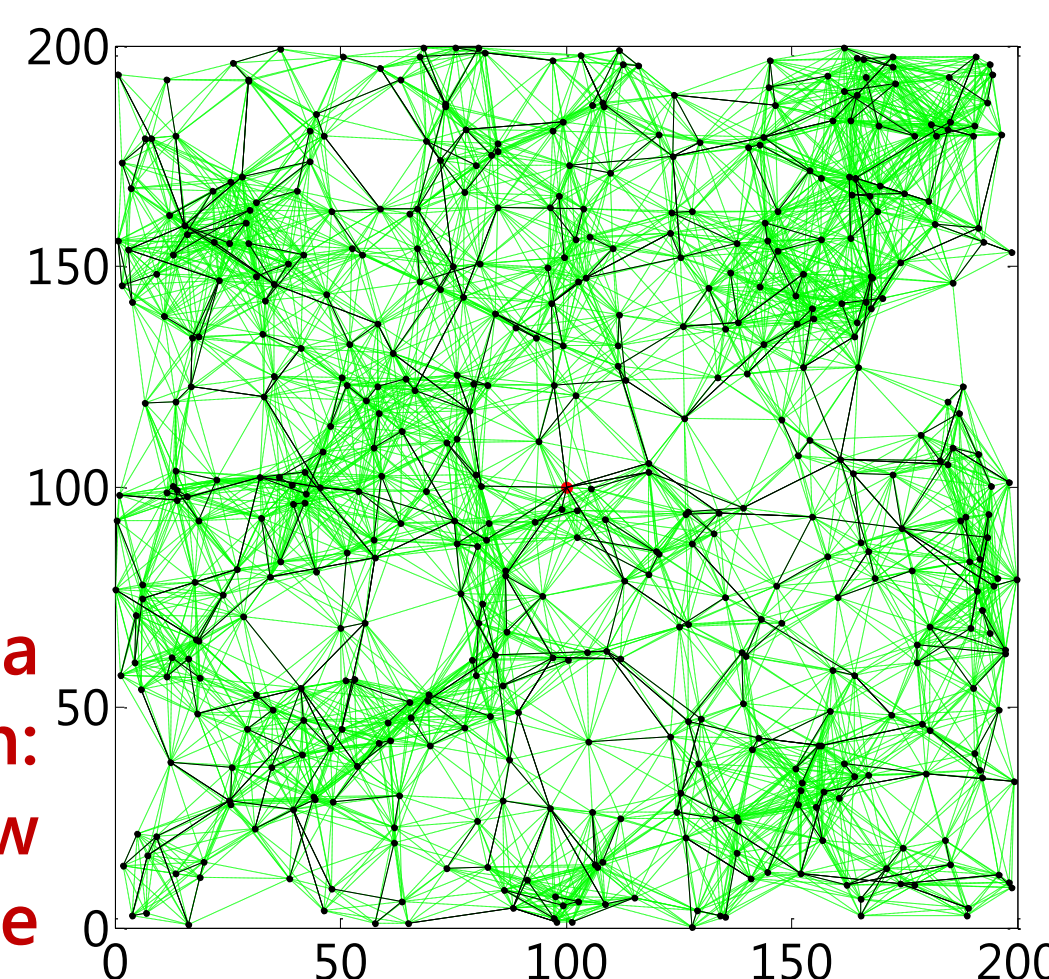
Approximation

- Design an (α, β) bi-criteria approximation to BDMRST on Unit Disk Graphs such that:
 - Maximum node degree in T is at most $\Delta^* + \alpha$
 - Radius of T is at most β times the optimal radius



Backbone Tree: Local roots $r_1, r_2, r_3, r_4, r_5, r_6$ are connected from non-empty adjacent cells (shaded) of sink s . Nodes r_3 and r_5 are connected using helper node w_1 (blue), and r_4 using helper node w_2 .

BDMRST on a 500 node graph: small radius, low degree



Approximation Algorithm ($\mathcal{A}$) for BDMRST

Phase 1: Backbone Tree (T_B)

- Tessellate the region into hexagonal grid cells of length $R/2$
- Choose one local root arbitrarily from each non-empty cell
- Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the set of local roots
- Connect the local roots in a **Breadth-First-Search** order starting from the sink s (using **helper nodes** when direct links do not exist), while ensuring the radius is not too long compared to an SPT on $\mathcal{R}$

Phase 2: Local Spanning Trees (T_j)

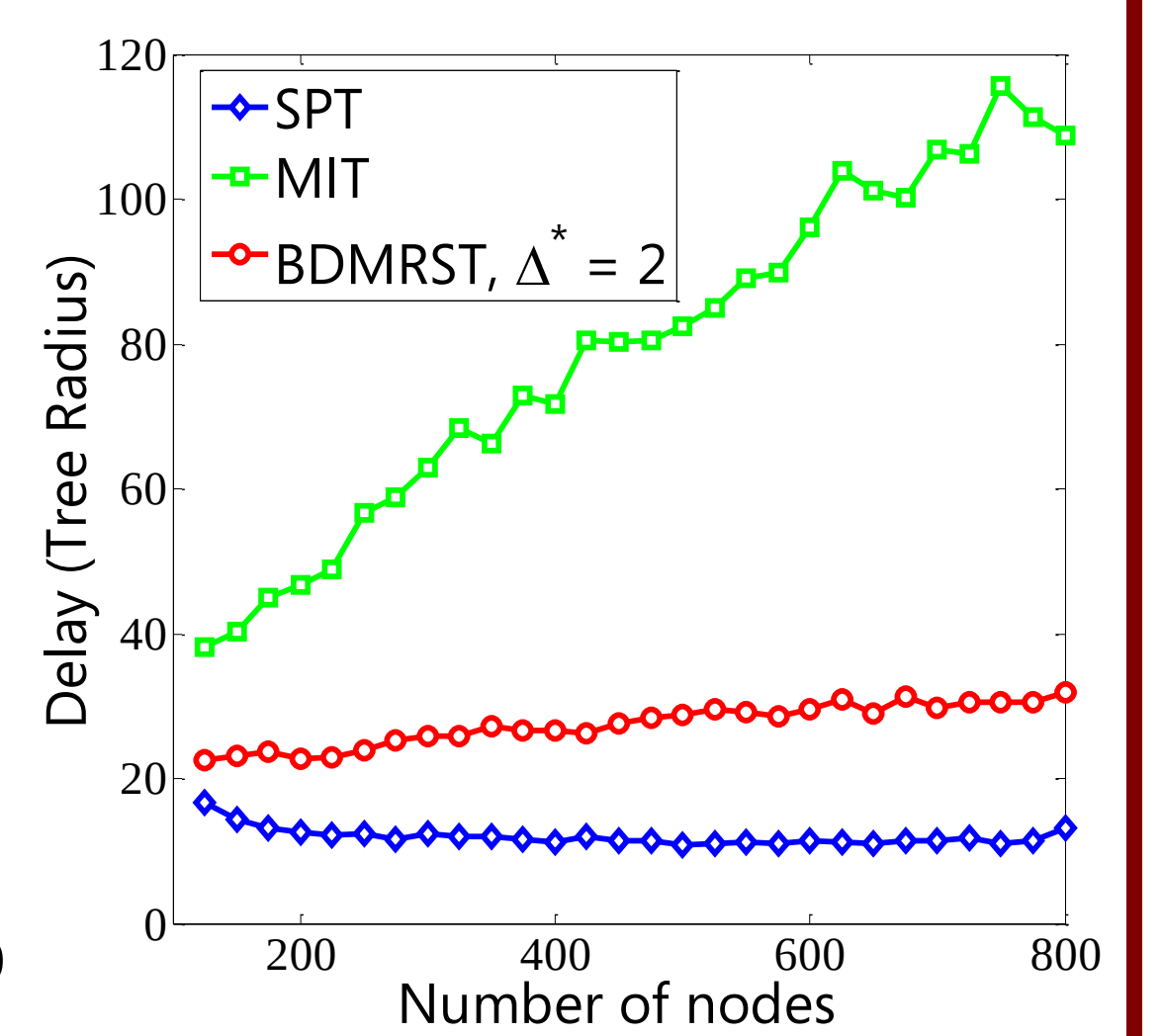
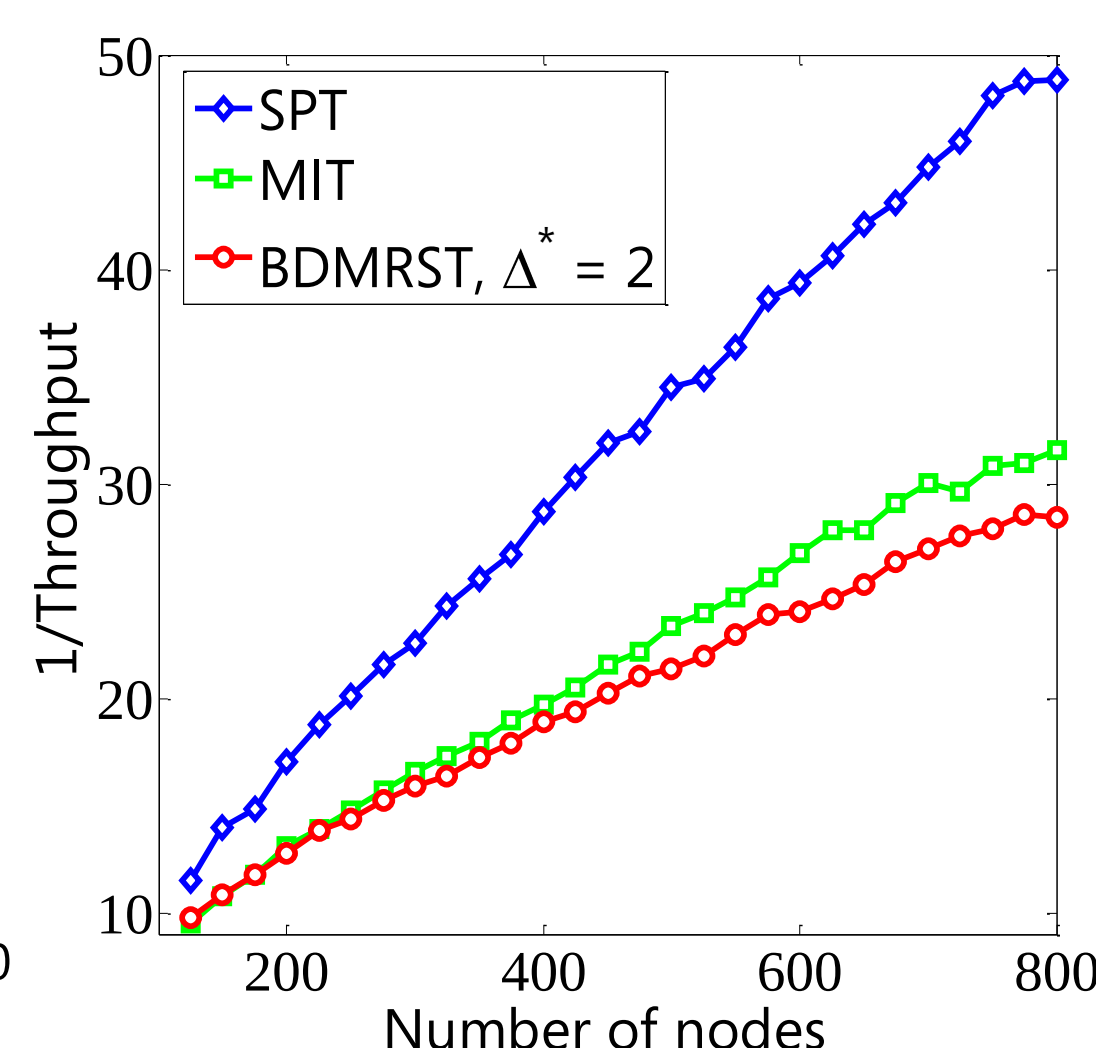
- Nodes within each hexagonal cell form a **complete graph**
- Construct local spanning trees T_j with the remaining nodes in each cell such that no node exceeds degree Δ^*

$$\text{BDMRST} = T_B \cup \{T_j\}$$

Main Result and Evaluation

Theorem

Algorithm $\mathcal{A}$ gives an (α, β) **constant factor approximation** to the Bounded-Degree Minimum-Radius Spanning Tree problem on Unit Disk Graphs, where $\alpha = 10$ and $\beta = 6$.



Evaluation of the interference-aware link scheduling algorithm in [1] on different trees. BDMRST achieves the best trade-off in terms of maximizing the aggregated sink throughput and minimizing the maximum delay.